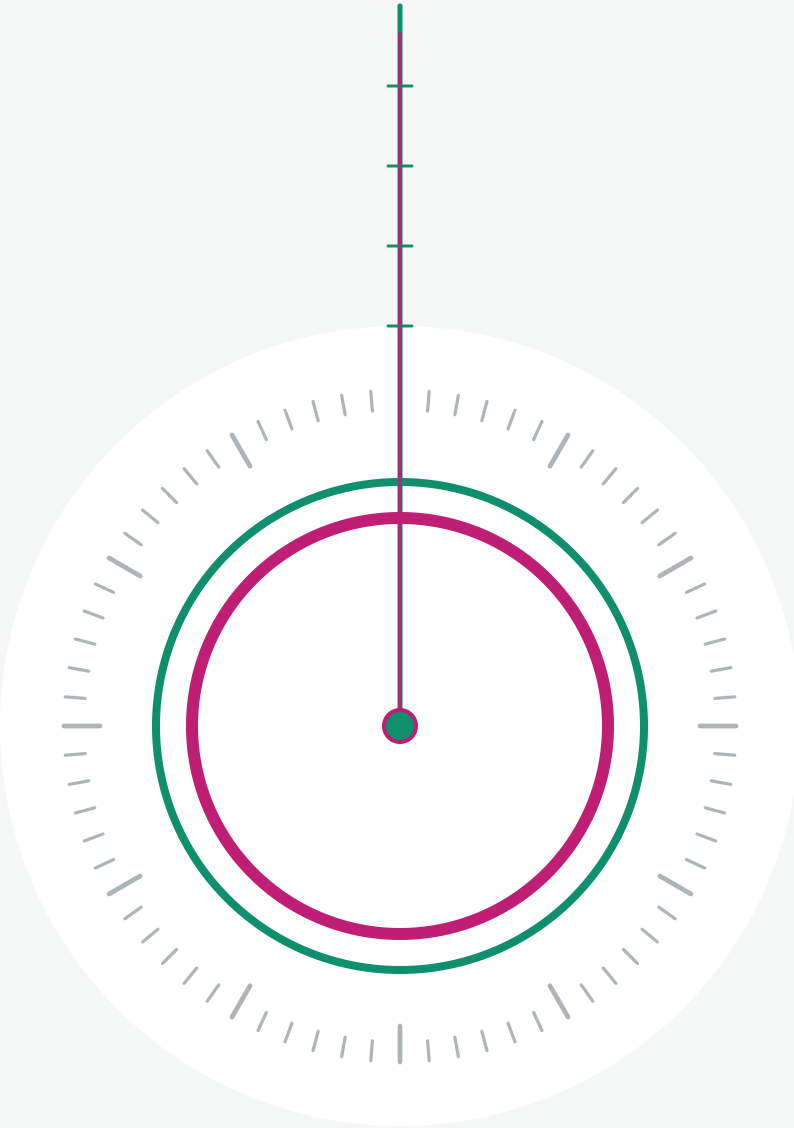
DATUM LINE

Every lab's cells. One "normal".

Membership to a shared repository
of every lab's control cells.



Kat Titterton
Cofounder, CSO, San Francisco



resume ↗

8
peer-reviewed papers
two in Nature Communications ↗

2
patent applications
filed at Axiom Bio

100,000+
compounds imaged
two Cell Painting datasets ↗

Kat Titterton

WORK

2023–2026 Axiom
cell imaging, AI

2022–2023 insitro
automated screening

2021–2022 MIT
expansion microscopy

2018–2021 abbvie
neuron assays

2017–2018 MGH
electrophysiology

EDUCATION

2013–2017 COLGATE BA
neuroscience, math

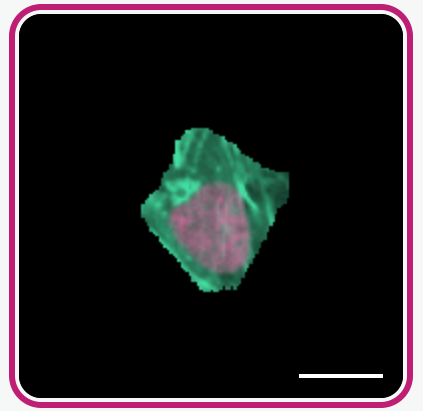
MY WORK

1. Cell Painting in primary human hepatocytes
   Cell Systems 2026, Ewald, Broad Institute
2. Counting cells predicts bioactivity benchmarks
   Nat Commun 2026, Seal, Univ. of Cambridge
3. Cell Painting of hepatotoxic botanicals
   bioRxiv 2026, Ollitrault, EMBL-EBI
4. An antibody to pathological tau
   Mol Neurodegen Adv 2026, Yanamandra, AbbVie
5. Capturing synaptic proteopathic seeds
   iScience 2026, Sebastian, AbbVie
6. A neuronal tau aggregation assay
   J Biol Chem 2025, Wu, AbbVie
7. Multiplexed imaging of brain nanostructures
   Nat Commun 2024, Kang, MIT
8. Tau aggregation in human iPSC neurons
   iScience 2022, Manos, AbbVie
9. Neurosteroid photolabeling of $GABA_A$ receptors
   Eur J Med Chem 2019, Wu, U Illinois Chicago

Do my cells look “normal”?

“normal” = published images + past data

asked @ five benches
… answered @ none



my cells, lab 2

DMSO, 20 µm bar

“Normal” compared with whom?
Each lab’s own checks can pass while labs still disagree (slides 03 and 04, archive A13).

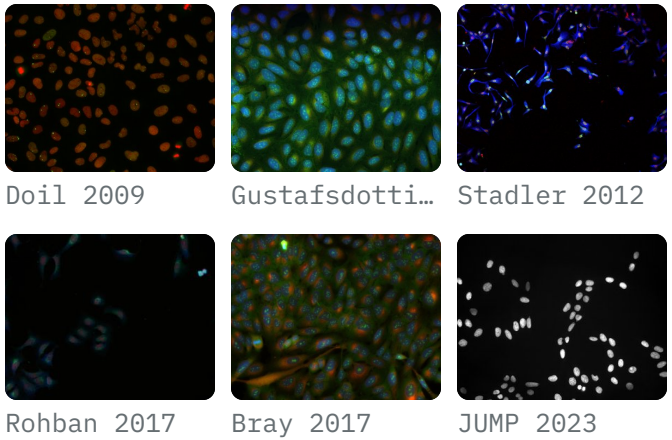
Today: search, ask, guess.

$28B a year of US lab research others cannot reproduce [6]

36% ≈ $10B, cells and reagents | 64%, other causes

search

U2OS cells “normal”



Doil 2009 | Gustafsdotti… | Stadler 2012

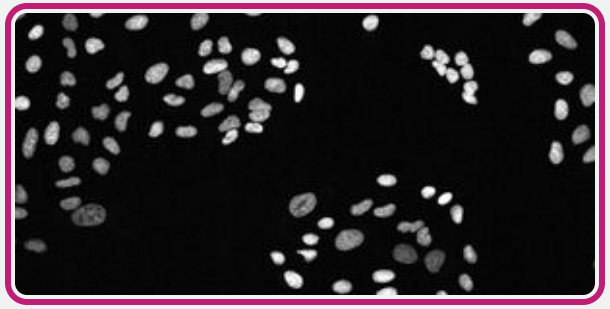
Rohban 2017 | Bray 2017 | JUMP 2023

six papers, six “normals”

ask

To: the paper’s authors

do my cells look right?



screenshot.png, Hoechst (DNA), lab 2

one opinion

guess

RUN IT

$100,000s

at risk

[28]: about $50k a run, 3 runs a week, in the author’s experience

WAIT

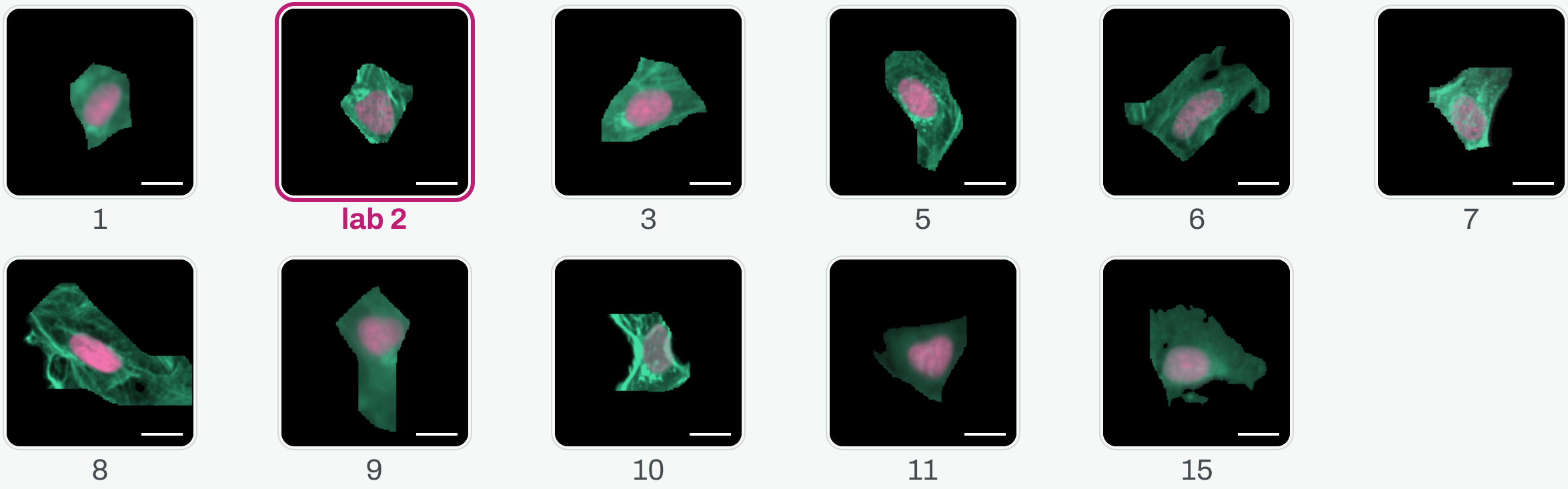
months

lost

[28]: asking around or re-running to be sure, the author’s estimate

search results: negative-control wells, IDR idr0010, idr0036, idr0025, idr0033, idr0016 (results/search-tiles.json); JUMP [7]; Doil and Rohban CC BY 4.0, Gustafsdottir and Bray CC0, Stadler CC BY-SA 3.0; costs [28]

Each lab judges “normal” alone.



nuclei (DNA) cell body (actin, membrane) — 20 µm

WHY NOW

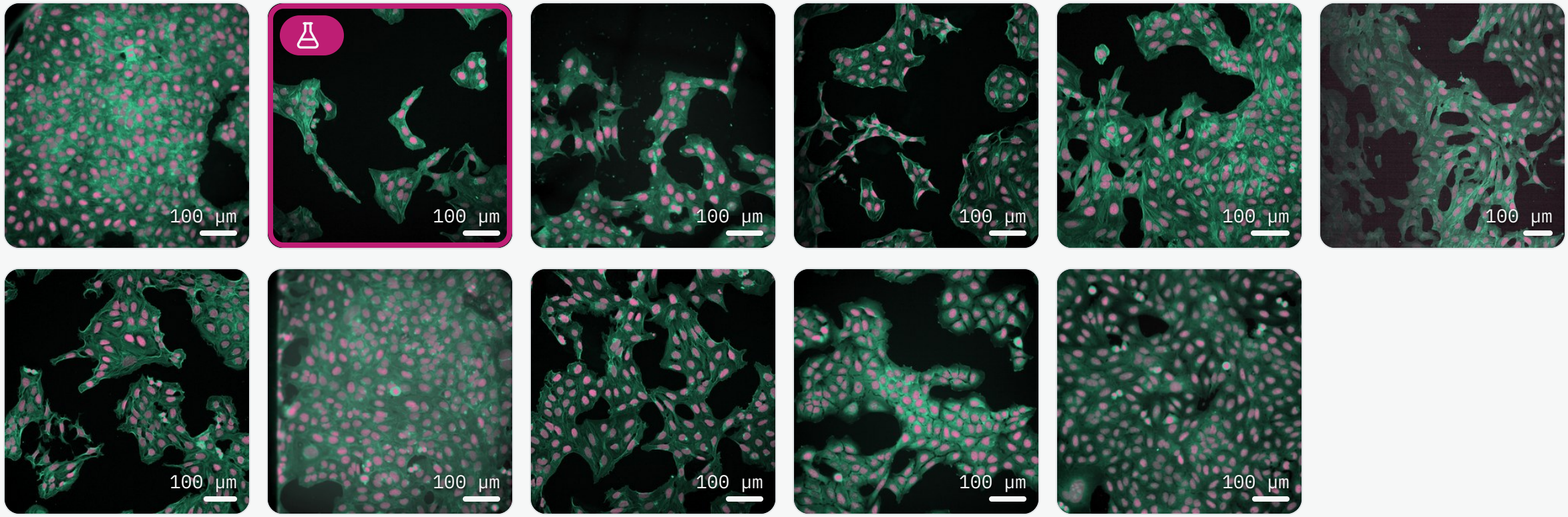
2023 1.8B public cells

2025 $30M per cell map

now AI reads cells

why now: 1.8B public cells, JUMP [7, 25]; FDA: fewer animal tests [9]; $30M per cell map, Roche to Recursion, Q3 2025 results [11]; AI reads cells (A3)

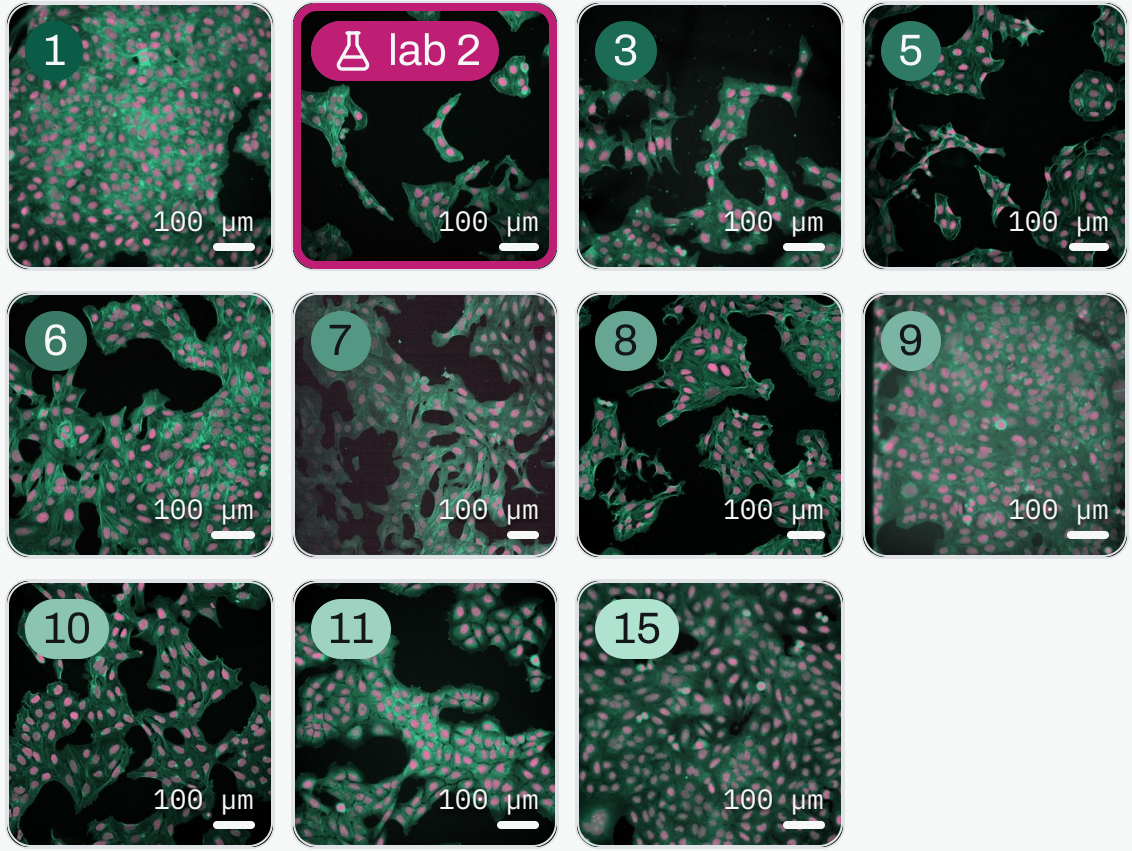
One lab, or eleven?
Guess out loud.



nuclei (DNA) cell body (actin, membrane) ✓ same cells ✓ same protocol ✓ no drug

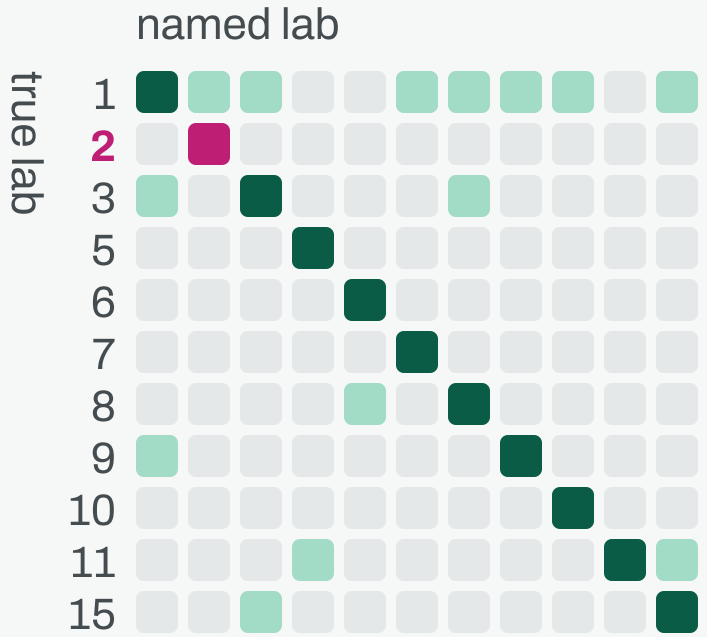
fig 1 Cell Painting, images: DNA channel, classifier: 3,773 numbers per well from all channels, 93,228 wells, 11 labs (a lab = one JUMP data source, its data-generating centre), plates held out, bars 100 µm, JUMP [7], archive A2, A1, archive A5

Could anyone tell?
A classifier can.



nuclei (DNA) cell body (actin, membrane)

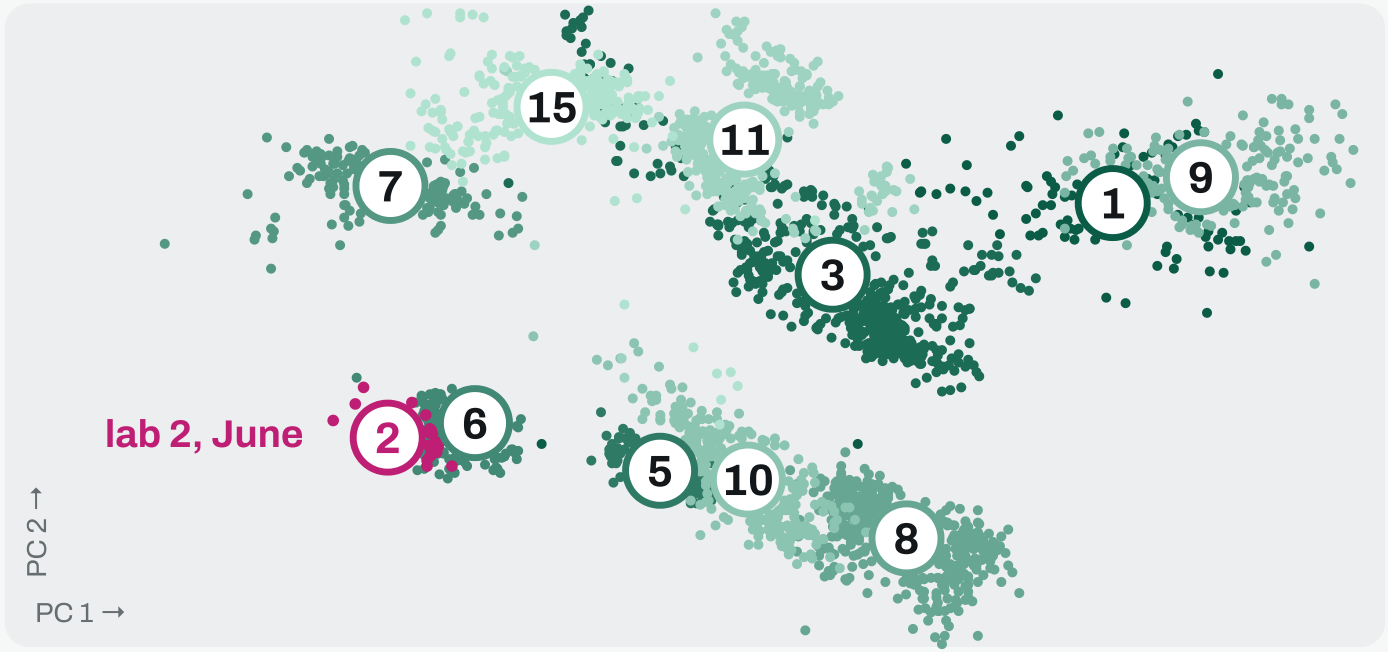
99.9% right lab, 93,228 wells
chance: 1 in 11



shade: the share of a row's wells, light grey none to dark all; 93,228 wells, results/e1

fig 1 Cell Painting, images: DNA channel, classifier: 3,773 numbers per well from all channels, 93,228 wells, 11 labs (a lab = one JUMP data source, its data-generating centre), plates held out, bars 100 µm, JUMP [7], archive A2, A1, archive A5

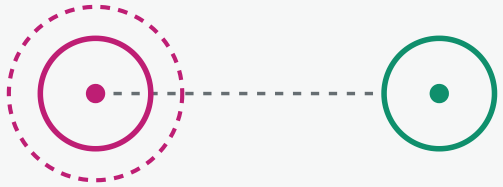
Eleven labs.
Eleven “normals”.



one well (15,352) 7 a lab’s “normal” closer = more alike

6×

between two labs



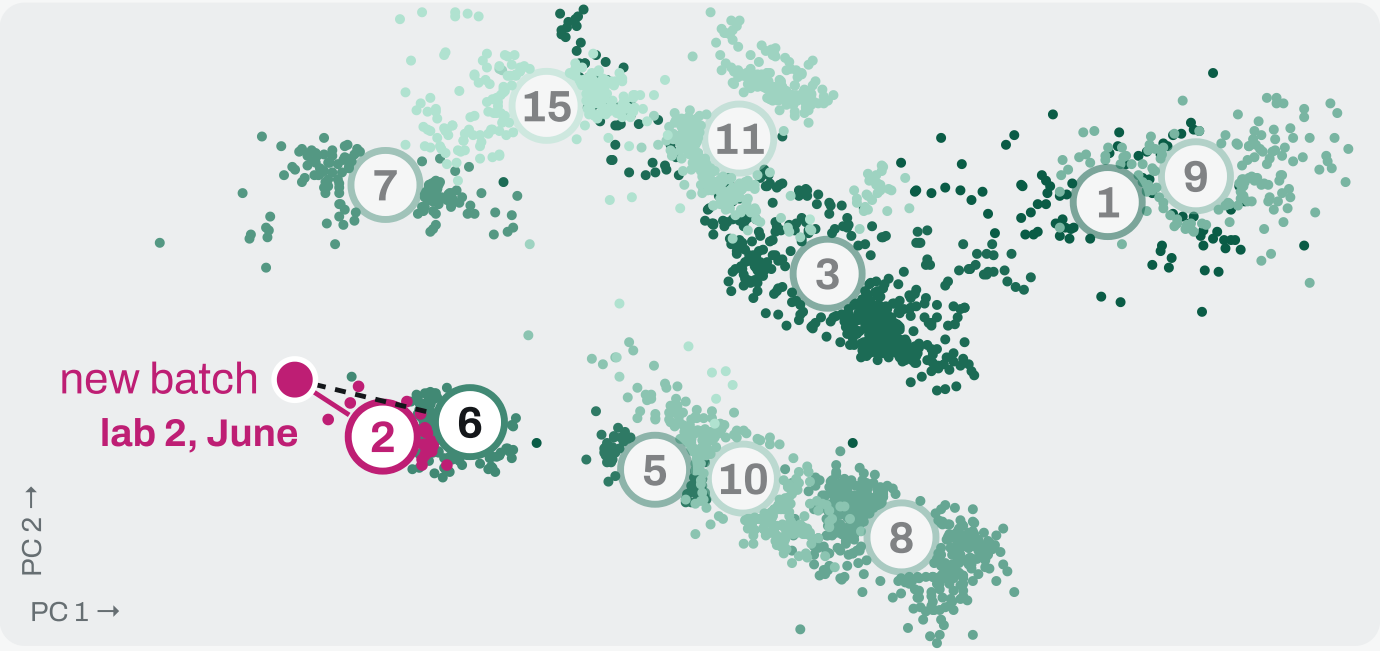
lab 2 another lab

- one well: ~220 cells
- 1× wells on one plate
- 1.6× a lab’s batches

fig 2 dots: DMSO wells, 8 per plate, one in three drawn; PCA of 15,352 wells, for the eye (distances use all 3,773 numbers); plates: 384 wells, 1,536 in labs 1 and 9; 6×: two labs’ centres sit 6.2 plate spreads apart on average, a plate spread being a well’s mean distance from its own plate’s centre, all 3,773 numbers (results/e13); JUMP [7]; archive A2, A1, archive A13

white paper ↗

Send the controls.
Know it’s “normal” before the screen.



one well 7 a lab’s “normal” new batch, 25 plates

Member check

a new batch, in all features

own lab	**2.8×**
next lab	**4.4×**

bar: distance, in plate spreads

- nearer its own lab ✓
- drugs match ✓
- grew as usual ✓

fig 3 the member check, replayed on public data; closest lab in all 3,773 numbers: lab 6, 4.4 within-plate spreads away; known drugs worked; agreement with other labs 0.70 measured, 0.77 on the map; 188 cells per well, in its usual range; JUMP [7], results/e9, e11, e12; archive A11, A4

white paper ↗

Controls catch drift
before any drug is read.

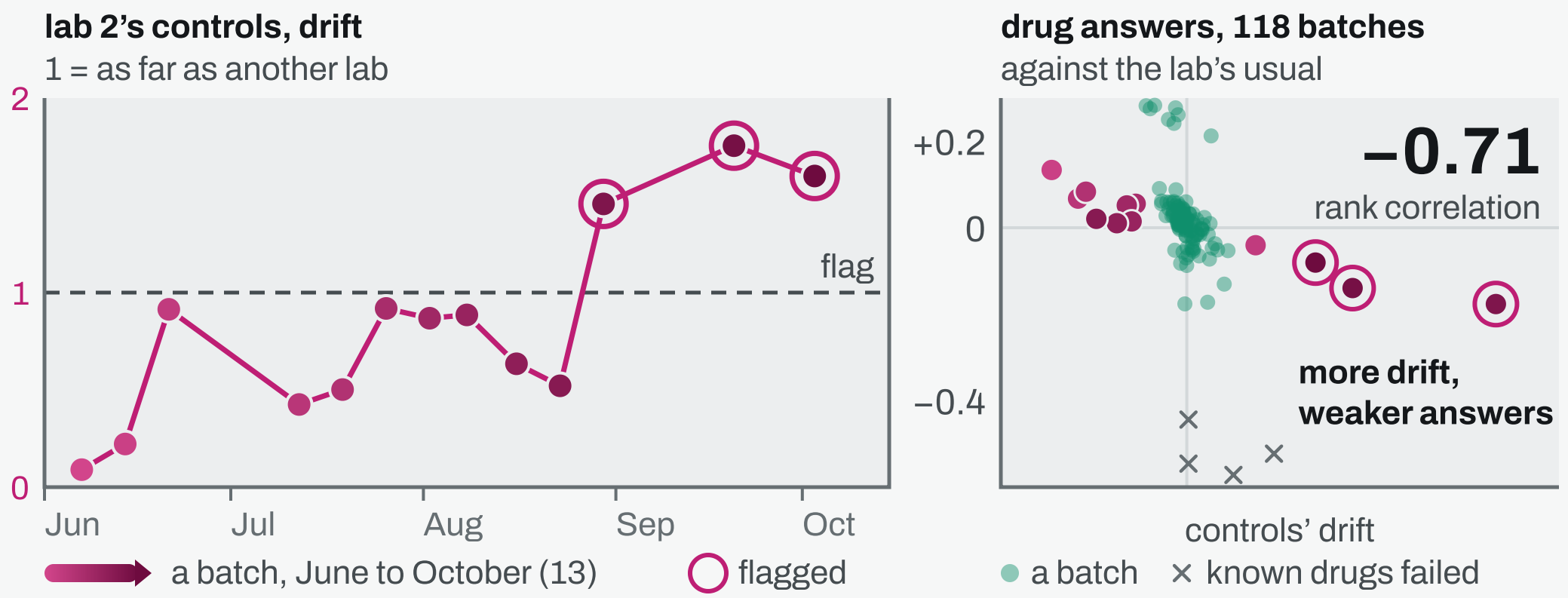
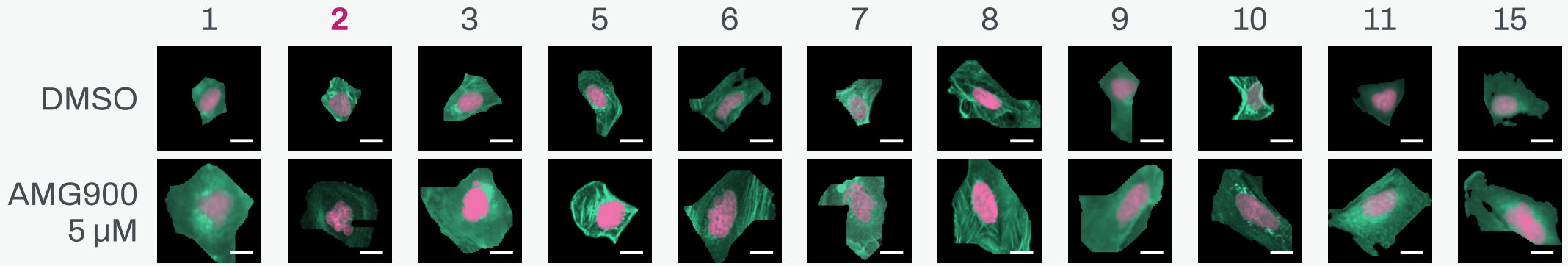


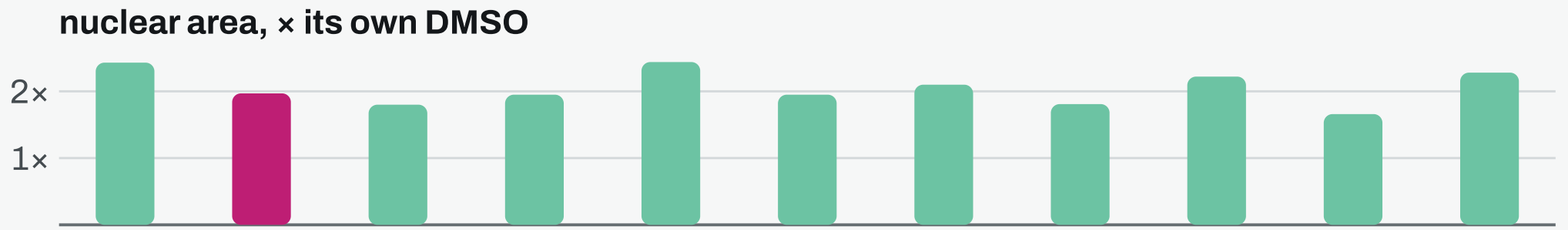
fig 4 lab 2, 13 batches; drift: control wells' distance from the first batch, in within-plate spreads; drift: distance from the first batch over distance to the nearest other lab's centre, so 1 is the flag line (most plates nearer another lab than the first batch, e9); right: every batch's drift and match minus its lab's mean, Spearman over all labs, permutation p = 0.0005, crosses: positive controls failed (results/e10); archive A6; JUMP [7]

white paper ↗

The product is the data.

One drug: nuclei ×1.7 to 2.4 in all 11 labs.

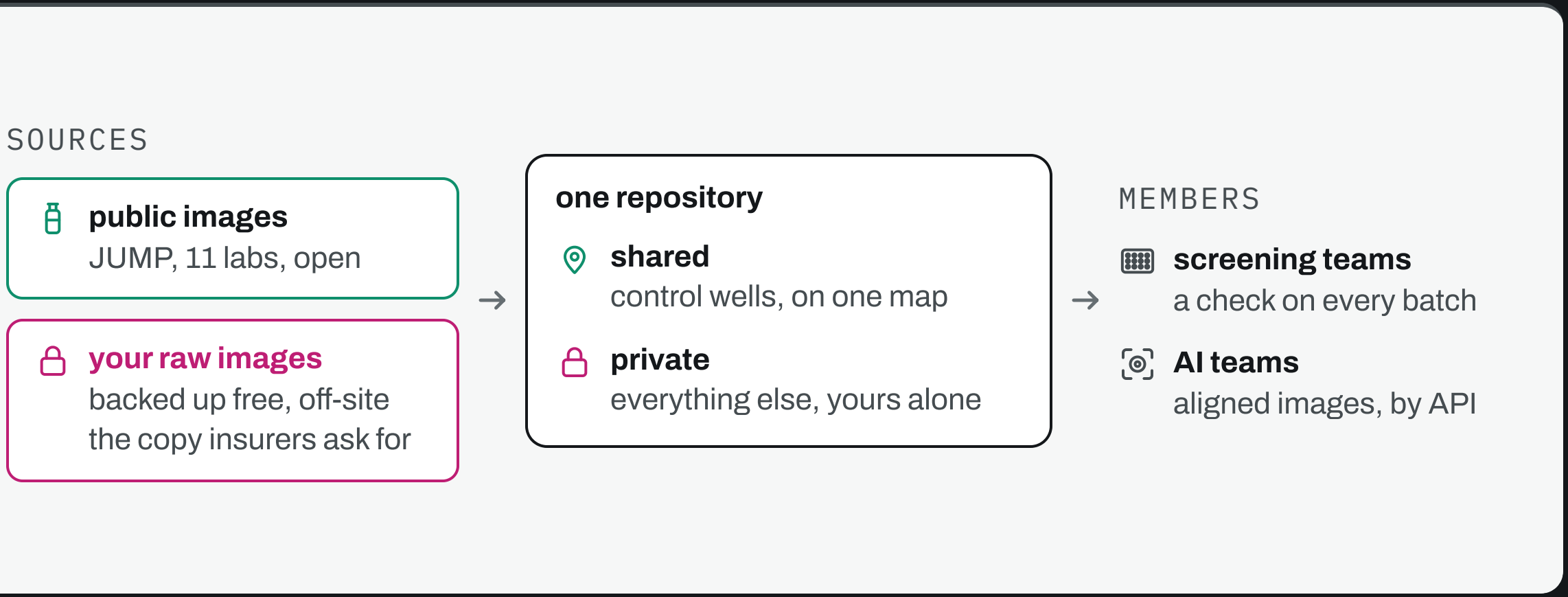




AMG900, an Aurora kinase inhibitor (cells copy their DNA but cannot divide), JUMP's strongest positive control, 5 µM; per lab, three plates whose known answer came out; each crop the cell of median nuclear area, 90 µm across; DNA pink, AGP green; bars 20 µm; one bar a lab, the median of 60 to 195 nuclei; mean ×2.1 (results/e18); archive A14; JUMP [7]

product vision ↗

11 labs' public images, plus yours. Only control wells are shared.



Proposed, except the public map (JUMP, CC0 [7]). Raw images backed up free, off-site and immutable: cold storage costs about $1 per TB a month [36]; one cyber insurer asks for a weekly offline or immutable backup [37]. Only control wells are shared, by choice; the rest is encrypted and read by no one else. A4, A9

product vision ↗

Every lab that joins sharpens “normal”: error 6.2 with one lab, 2.8 with five.

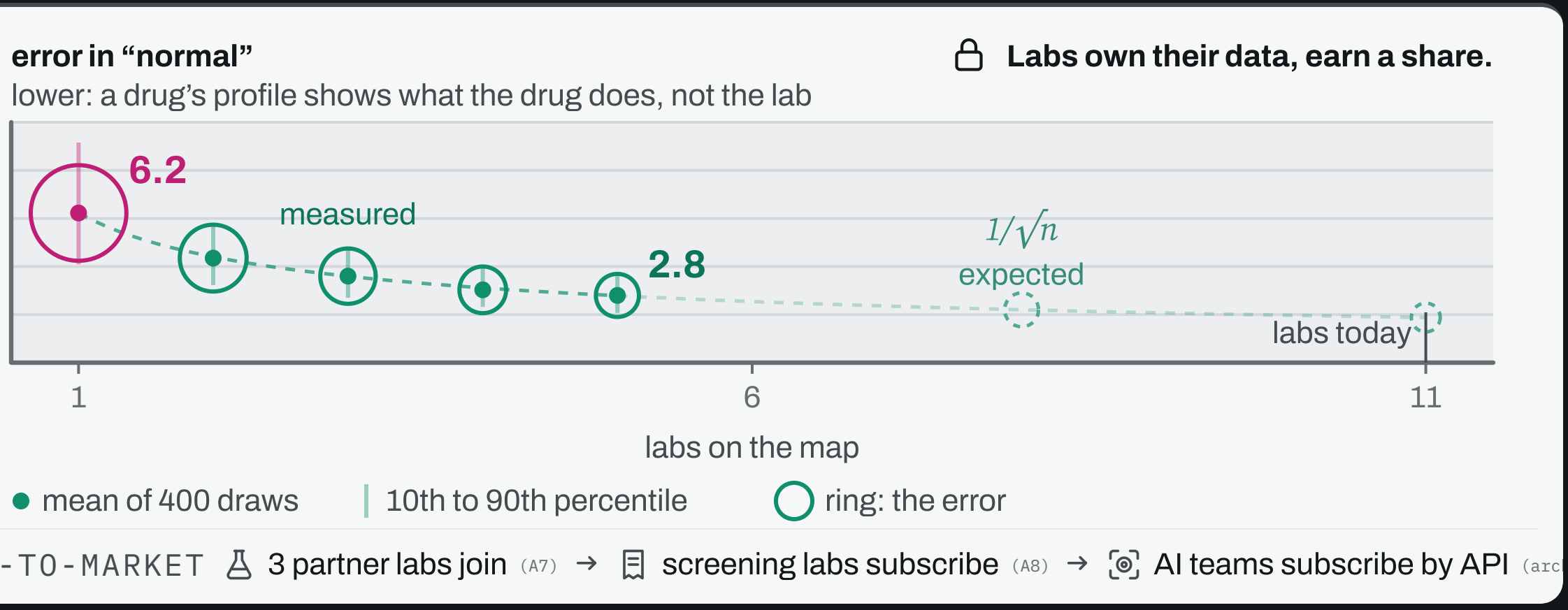
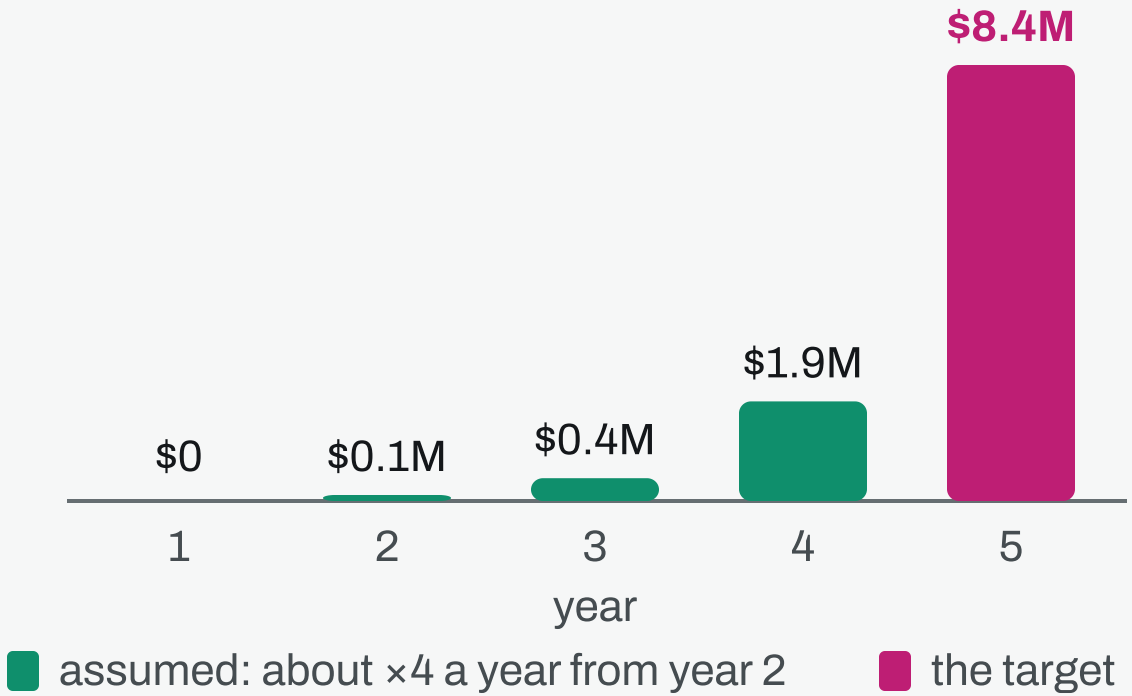


fig 5 error: distance between two maps from different labs, in within-plate spreads; dot: mean of 400 draws, bar: 10th to 90th percentile; measured to five labs, dashed beyond; 1,871 plates; JUMP [7]; results/e13; why: RSL3 at 10 µM agrees 0.90 inside a lab, 0.49 across (A2, e16)

Every likely buyer: $84M/year.
Five-year target: a tenth, $8.4M/year.

five-year plan, revenue a year



who pays, every likely buyer



a tenth: $8.4M/year

data contributors	free access	$0
screening labs	$10k/line/quarter	$64M
buyers of cell data	$25k/dataset	$15M
big screening panels	$150k/year	$3.75M
benchmark partners	$100k/year	$1M

fig 6 base case of A8: every count and price an assumption; low case $9.7M/year; data contributors join free and are the map, not the revenue; AI usage by API not priced; beyond the base case, AI cell data is a $1.8B push [12] and irreproducible research costs $28B/year [6]

Each tool sees one lab.
Datum Line sees every lab.

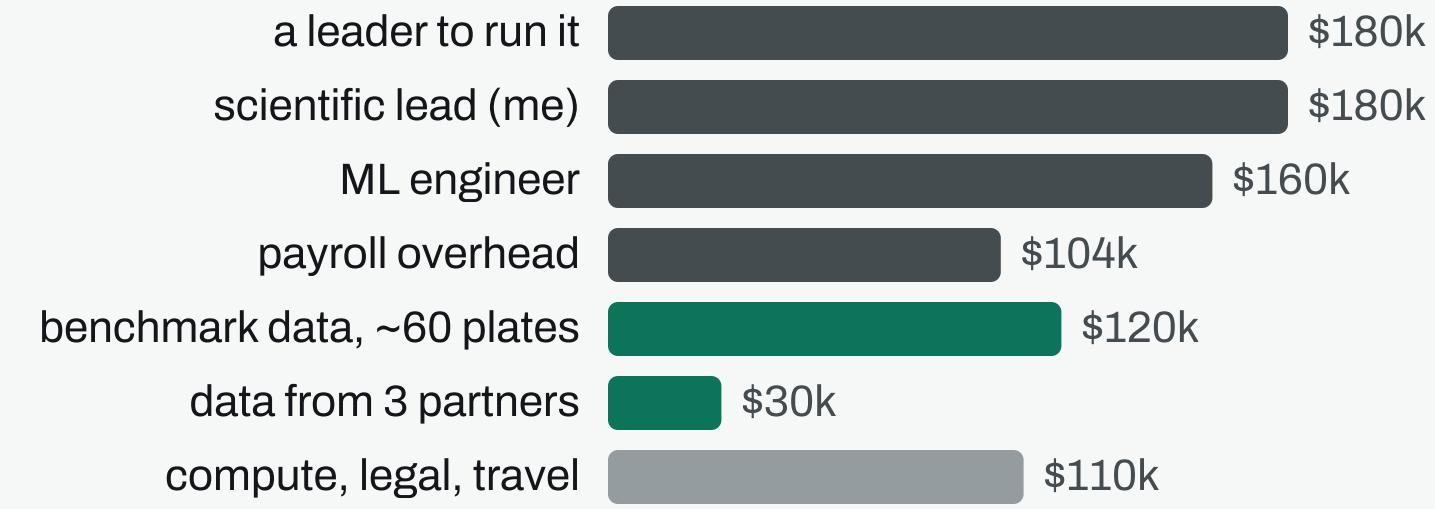
	measure cells	normalise plates	reference data	every lab, over time
CellProfiler image analysis, open source [32]	✓			
Harmony Revvity's Opera Phenix software [33]	✓			
pycytominer profiles, each plate to its own DMSO [34]		✓		
Genedata Screener screen QC and normalisation [35]		✓		
JUMP, Recursion reference data at scale [7, 11]			✓	
Datum Line every lab's controls, one repository				✓

Datum Line hosts every lab's raw control images, and runs on them.

tools as a phenotypic screen runs them: CellProfiler [32]; Harmony, Revvity [33]; pycytominer [34]; Genedata Screener [35]; JUMP [7], Recursion [11]. A check: the tool does the step, from its own documentation. The wider landscape (cell-line checks, vendor certificates): docs/landscape.html

landscape ↗

Raising $1.8M pre-seed for 18 months: first paying members, then the seed.



$884k plan + $884k contingency = $1.77M (A7)

TODAY

- **11 labs, 129 batches mapped** from public data (JUMP [7], results/e9); no revenue yet

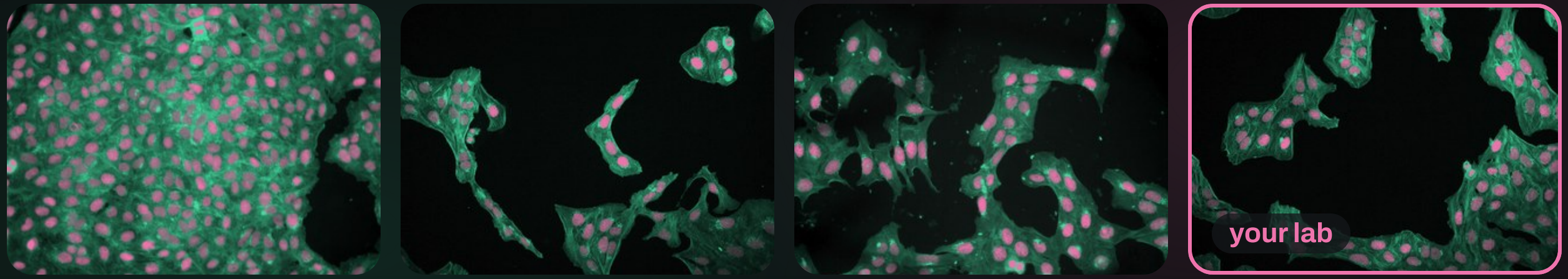
BY MONTH 18

1. **data from three partner labs, the first outside JUMP**
2. **first paying member**
3. **benchmark playground**
4. **model for any cell**

→ seed round

proposed; milestones in order, not dated; every line an assumption (A7)

roadmap ↗



There's a tile for your lab.
Let's map "normal" together.

$1.8M
pre-seed for 18 months

Have control images? Talk to me.
kat-titter.github.io/datum-line

pre-seed: a leader, three partner labs, two benchmark partners (A7); open code: github.com/kat-titter/datum-line

The rest, for questions.

Every number: the code here or a cited source.
All 25 boards: archive.html
Documents: white paper, product vision, roadmap, landscape

A1 **Why not batch-correct it away?**
erases most of the lab fingerprint

A2 **Does it change the result?**
same drug, different answer

A3 **Does it work without stains?**
bet on brightfield

A4 **What is private?**
control wells shared, nothing else is

A5 **What is the machine learning?**
image in, position and correction out

A6 **What about biosecurity?**
cells show their lab, labels can lie

A7 **How much, and what for?**
planned $0.9M × 2 = $1.8M asked

A8 **How big?**
count × price, three streams

A9 **Why would a lab share?**
owned by the lab, verified by anyone

The usual correction erases most of the lab's fingerprint.

JUMP records the left column; nobody records the middle.

RECORDED IN JUMP

- lab
- batch date
- plate
- plate type
- well position
- microscope
- confocal or widefield
- laser or LED
- objective NA
- fields per well
- filter set
- pixel size
- brightfield planes
- CellProfiler version

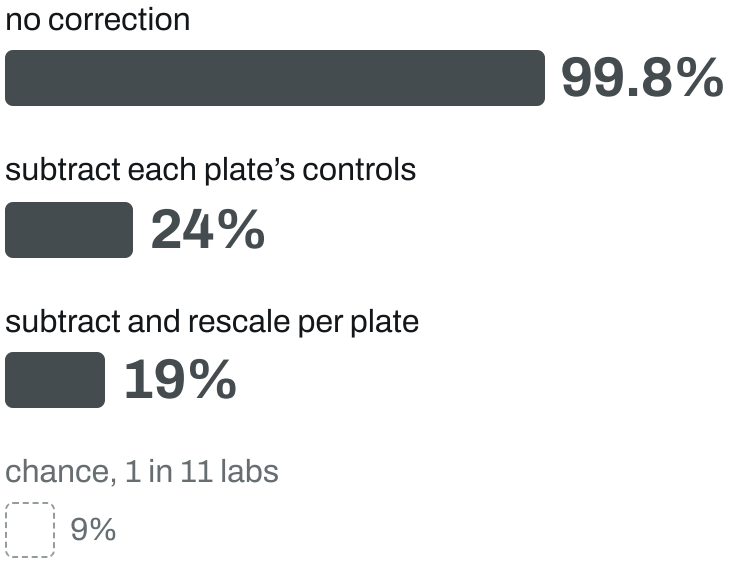
in every image file too:
image size, stored bit depth

NOT IN ITS PUBLIC METADATA

- DMSO % per well
- DMSO lot
- DMSO water uptake
- stock freeze–thaw
- serum lot
- media lot
- passage
- thaw date
- seeding density
- confluence at dosing
- dosing to fixation
- stain and dye lots
- operator
- incubator shelf
- edge evaporation
- mycoplasma status
- CO2 and humidity
- compound dose

not required to publish today
in our own data: tracked, every one

CAN A MODEL TELL WHICH LAB?
HIGHER = CLEARER LAB FINGERPRINT



29,943 DMSO wells, 16 per plate
tested on plates it never saw

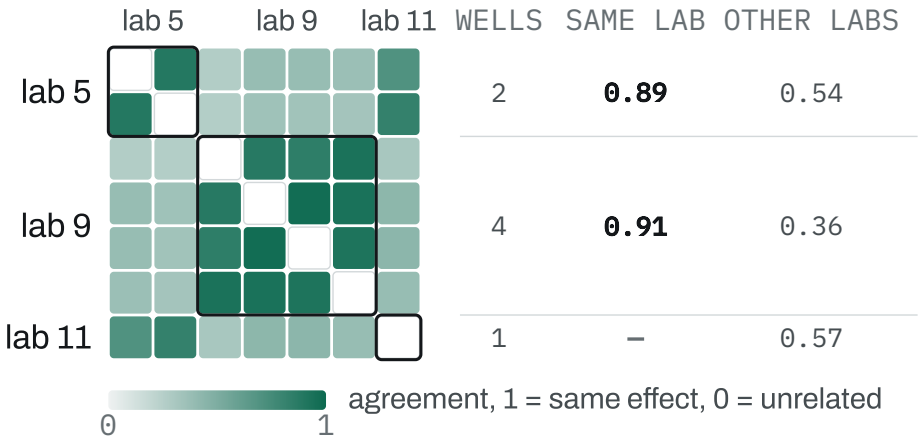
Batch correction is built to remove this signal [18], variables: JUMP metadata [7]

The same drug looks less alike in another lab.

Darker = more alike.

WHAT A DRUG SCREEN REPORTS drug wells – DMSO wells, same plate

RSL3 AT 10 µM, EVERY WELL AGAINST EVERY OTHER



	WELLS	SAME LAB	OTHER LABS
lab 5	2	**0.89**	0.54
lab 9	4	**0.91**	0.36
lab 11	1	–	0.57

SAME 7 PLATES, EACH WELL AGAINST ITS OWN PLATE'S DMSO

	SAME LAB	OTHER LABS
RSL3, 10 µM	**0.90**	0.49
8 known drugs, 5 µM, median	**0.74**	0.26
DMSO vs DMSO	**0.00**	0.00

DOES IT CHANGE THE NUMBER?

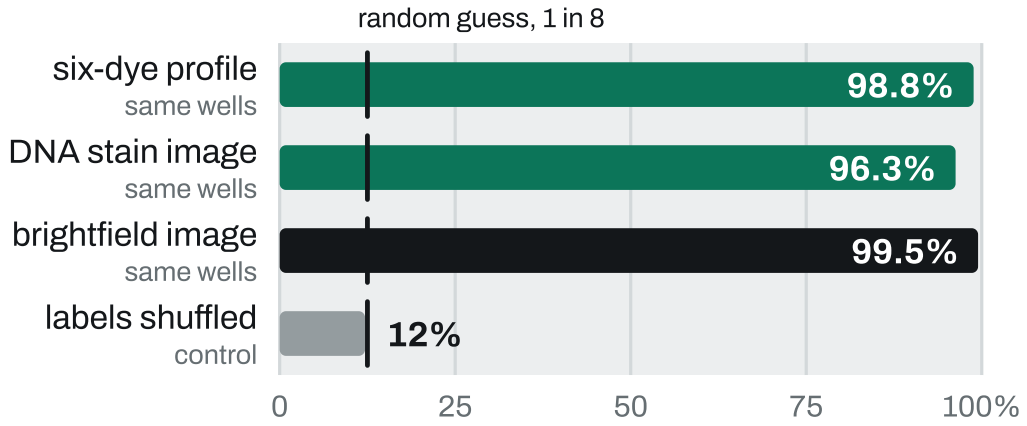
Which lab ran it explains 20% to 71% of how strong a drug's effect looks.

Seven wells: an example, not proof, lab 7, dosed at a sixteenth, left out, agreement = cosine of two wells' effects, each well minus its own plate's DMSO wells, 8 drugs at full scale: 1,829 plates, 11 labs, JUMP [7]

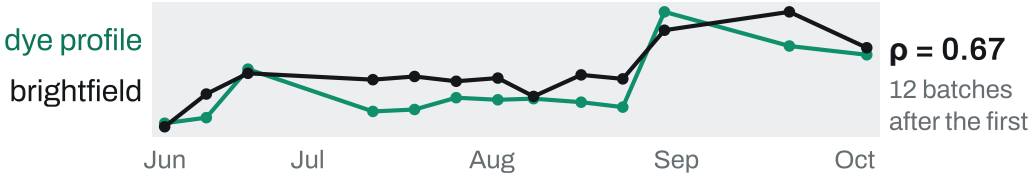
One brightfield image already names the lab.

Evidence: six dyes.
Product: no dye.

CAN ONE WELL TELL WHICH OF 8 LABS IT CAME FROM?



DOES BRIGHTFIELD SEE LAB 2 DRIFT?, EACH LINE SCALED TO ITS PEAK

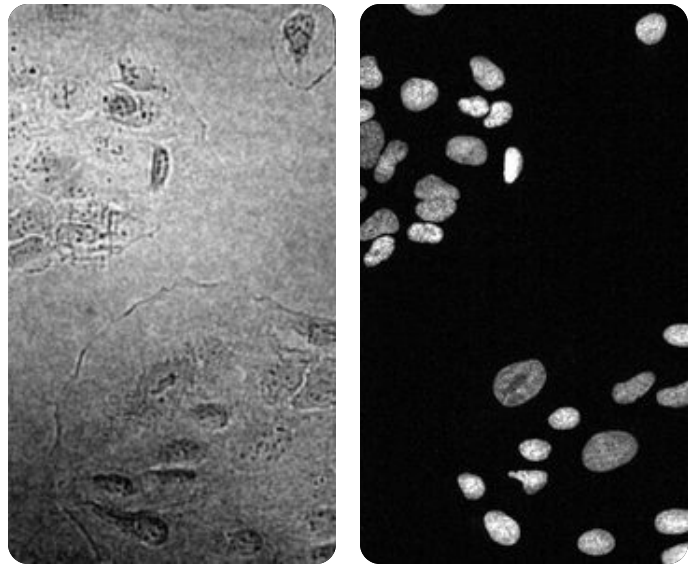


Not yet the whole map: beyond each lab's usual neighbour, brightfield and dye profile agree batch by batch no more than luck (45 of 102 against 41).

WHY THE BET IS FAIR

No dye: fewer things can go wrong

~~stain lot~~ ~~dye lot~~ ~~staining step~~ A1



BRIGHTFIELD DNA STAIN

same spot in one well, lab 2

% named right, each lab counted equally, unseen batches, one DMSO well per plate, 1,338 plates, the same wells and model for every row, median of 20 splits of the batches, profile without its 8 frame-size columns, shuffled: 5 seeds per input, image model never trained on cells, JUMP [7]

Only control wells are shared.

A free backup of the rest is yours alone.

WHAT IS SHARED

control wells only: DMSO and known drugs



green: 64 of 384 wells shared, JUMP [7]

images, or numbers from them

proposed: images deleted once measured

a plate map and the batch date

which wells, and when they were imaged

WHAT DATUM LINE DOES WITH IT

builds the map

scale set by the other labs, never its own

the map holds DMSO wells only

82,319

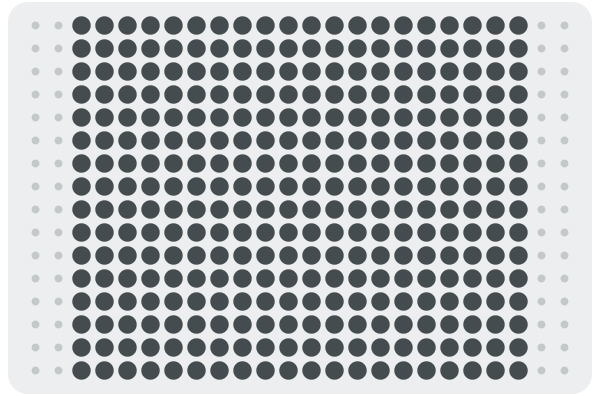
for lab 2: DMSO wells of 10 other labs
0 treated

answers

“normal” or not, how far it moved, nearest lab

WHAT IT NEVER READS

your treated wells



dark: 320 of 384 wells stay yours

your results

corrected in your building, by you

your name, unless you choose

proposed: a number until you opt in

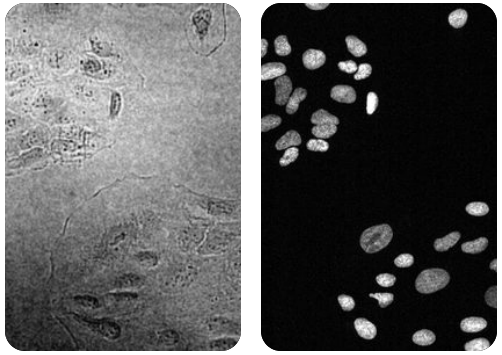
your backup

proposed, free: encrypted with your key

In every analysis here, the map and the correction see DMSO wells only, github.com/kat-titter/datum-line/analysis, JUMP [7]

An image goes in; a position and a correction come out.

1, AN IMAGE



brightfield DNA stain

one DMSO well, JUMP [7]

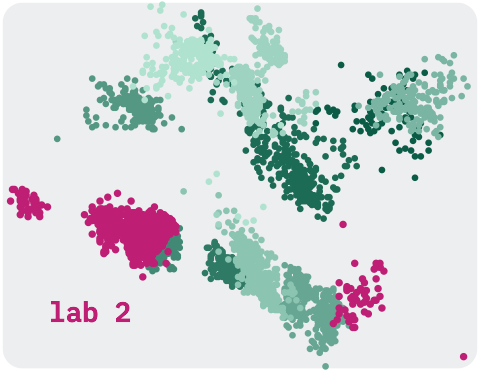
2, NUMBERS

3,773

numbers per well

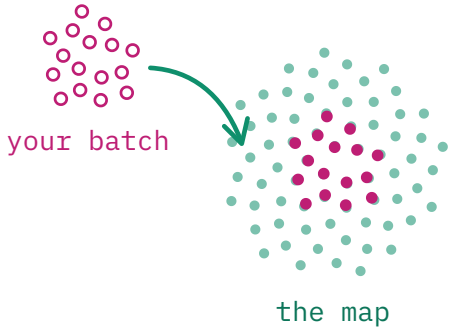
standard software today: size, shape, brightness, texture

3, THE MAP



each lab placed on the other labs' scale, never its own

4, THE CORRECTION



one per batch, fitted on your DMSO wells, sketch, archive A14

AI IMAGE MODEL NEXT, A3



right lab, from one picture

Correction, per batch on 200 components: $x' = C_{field}^{\frac{1}{2}} C_{batch}^{-\frac{1}{2}} (x - m_{batch})$, every script: github.com/kat-titter/datum-line/analysis

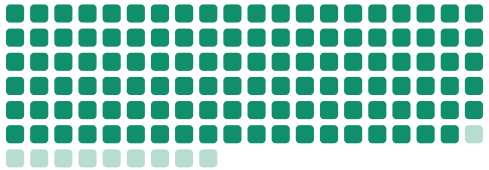
Every lab leaves a fingerprint, so a wrong label stands out.

Two work today, three are proposed.

PROVENANCE today

99.9%

of 93,228 DMSO wells get the right lab, on unseen plates: cells and microscope together.



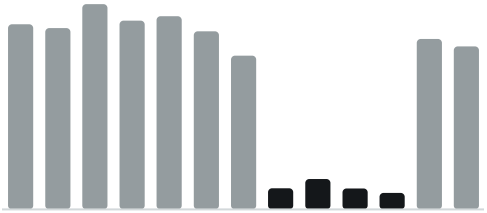
1 square = 1 batch
solid: every well right, 119

INTEGRITY today

4 batches

Label: known drug in these wells.

Cells: almost no effect, on 65 plates.



lab 3, drug effect per batch
dark: failed, archive A8

UNKNOWN ORIGIN proposed

9×

Farthest of 129 real batches from its nearest other lab.

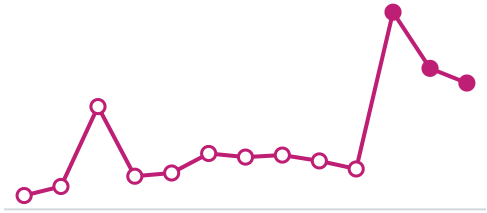


distance to nearest other lab
1 line = 1 real batch

EARLY WARNING proposed

30 Aug

2021: lab 2's DMSO wells moved; its cell count did not.



lab 2, since batch 1
filled: flagged, archive A13

DECLARED VS OBSERVED

proposed, opt-in

A company that wants the good-actor mark sends treated wells and says what is in them. The map flags cells that look like a known toxic class [7]. Only what a lab sends is seen.

Integrity check uses each plate's known-drug wells (archive A8), × = times the usual spread of wells within one plate, JUMP [7]

proposed, every line an assumption

About $1.8M for eighteen months, mostly people.

$884k plan + $884k contingency = $1.77M asked



people 71% | benchmark 17% | other 12%

leader, to run it	18 months	**$180k**
scientific lead (me)	18 months	**$180k**
ML engineer	from month 7	**$160k**
payroll overhead	20% of salaries	**$104k**
benchmark playground	~60 plates, planted faults, archive A16	**$120k**
partner-lab bench time	3 labs, controls they already image, reagent top-ups, shipping	**$30k**
compute and storage	images, models, hosting, API	**$45k**
legal, IP, insurance, accounting	incorporation to first contract	**$45k**
travel and partner visits	2 benchmark partners, 3 labs	**$20k**
planned	sum	**$884k**
+ contingency	most things take 2–3× the time and money planned, we plan for 2×	$884k
asked, 18 months		**$1.77M**

NO IN-HOUSE LAB

Partner labs or a contract lab.
Own lab: costs more, proves less.

WHAT THE MONEY MUST SHOW

1 data from three partner labs, the first outside JUMP, paid in this round **2** first paying member
3 benchmark playground **4** model for any cell → seed

salaries below market, leader and scientific lead paid, nothing here committed

About $84M a year, if every likely buyer joined.

assumption	low	base	high
organisations testing drugs on a few cell lines	200	400	800
cell lines each one uses per quarter	2	4	8
price per cell line per quarter, $k	5	10	15
organisations testing drugs on hundreds to thousands of cell lines	10	25	50
site licence per year, $k	50	150	300
per-line checks and site licences, per year	**$8.5M**	**$67.8M**	**$399M**
companies buying cell data they did not make	50	150	300
purchased datasets each places on the map per year	2	4	8
price per dataset membership, $k	10	25	50
dataset memberships on purchased data, per year	**$1M**	**$15M**	**$120M**
benchmark partners	4	10	20
membership per partner per year, $k	50	100	200
benchmark membership, per year	**$200k**	**$1M**	**$4M**
total per year, if every buyer joined	**$9.7M**	**$83.8M**	**$523M**

BASE CASE, PER YEAR



↑ five-year target, a tenth, $8.4M

benchmark $1M ↑

WHY BUYERS PAY

A line checked late, in a screen I ran:
3 runs a week × 17 weeks × $50k ≈ $2.6M to redo.
A $25k dataset membership: about 1% of that.

prices proposed on archive A15, academic labs join free, they are the map, not the revenue

The lab keeps its data and a share; anyone can check it.

No crypto token.

CONTROL WELLS ONLY — today

no test compound, no target, no result
is ever shared or read

LAB KEEPS ITS DATA — proposed

sharing is opt-in
withdraw any time

LOW RISK BY DESIGN — today

cell-line images only
no human genetic data, no pathogen data

SIGNED AND TIMESTAMPED — proposed

public record, anyone can check
dates cannot be changed, blockchain optional

KNOWN MEMBERS — proposed

verified at sign-up, odd access flagged
access log open to members

OPEN CODE AND DATA — today

code: MIT, on GitHub
data: JUMP, CC0; every number from a script

lab shares → **the map checks** → **members subscribe** → **lab keeps a share**

control images, opt-in
credited by name on the map

how “normal” the cells look
which lab, what changed

to aligned data
from every lab

of every membership
as long as its data is used